DO YOU KNOW?



Henry Maudslay

(22 August 1771 – 14 February 1831) was a British machine tool innovator, tool and die maker, and inventor. He is considered a founding father of machine tool technology. His inventions were an important foundation for the Industrial Revolution.

of shaft and hole, the zero line will be lying on the minimum diameter of the hole as shown figure. For this system the lower limit size of hole is equal to basic size.

SHAFT BASIC SYSTEM

If the system of assembly of shaft and hole consisting of basic shaft, then that type of system is known as shaft basic system. It means for the assembly of shaft and hole, the zero line will be lying on the maximum size of the shaft as shown figure. For this system the upper limit size of shaft is equal to the basic size.

LIMIT SYSTEM

Limit system is important to keep limits and fits within desired limits. It has to be standardized to be able to produce components with interchangeability.

There are several organisations worldwide to ensure standardization. They are

1. International Standardization for Organisation (ISO)
2. Newall system
3. Indian Standard Institution (ISI)

8.7 INTERNATIONAL STANDARD ORGANISATION (ISO)

ISO is an International Standard setting body composed of representatives from various national standards organisations. It was founded on 23 February 1947. The organisation promotes worldwide proprietary, Industrial and Commerical Standards.

It's headquarters located Geneva at Switzerland.

In this system, Setout in BS 4005 – 1969, allows for 27 types fits and 18 grades of tolerance covering a size ranging from 0 to 3150 mm. In this system 27 possible holes are designated by capital letters A, B, C, D, E, F, G, JS, J, K, M, N,O, P, R, S, T, U, V, X, Y, ZA, ZB and ZC and the shaft by lower case letters, a, b, c, d, e, f, g, js, j, k, m, n, o, p, r, s, t, u, v, x, y, za, zb and zc.

The 18 grades of tolerance are designated by numerals

T01, T0, T1, T2, T3, T4,T5, T6, T7, T8, T9, T10, T11, T12, T13, T14, T15 and T16

For specifying a particular hole or shaft, it is to be written as the letter followed by the numeral.

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